

Second Terminal Examination - 2079

Class : XI

Subject: Mathematics (0071)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The contrapositive statement of $\sim p \Rightarrow q$ is
 a) $\sim q \Rightarrow \sim p$ b) $q \Rightarrow p$ c) $\sim p \Rightarrow \sim q$ d) $\sim q \Rightarrow p$
- For any two sets A and B , $A - B =$
 a) $A \cap \bar{B}$ b) $\bar{A} \cap B$ c) $A \cup \bar{B}$ d) $B - A$
- If $A = (-\infty, 0)$ and $B = [0, \infty)$ then $A \cup B$ is
 a) $\{0\}$ b) $(-\infty, \infty)$ c) $(-\infty, \infty) - \{0\}$ d) $\{-\infty, \infty\}$
- Let $A = \{2, 3\}$, then a relation on A defined by $x + y < 4$ is
 a) $\{\}$ b) $\{(2, 2)\}$
 c) $\{(2, 3), (3, 2), (3, 3)\}$ d) $A \times A$
- If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ then $A \cdot A^T$ is
 a) 0 b) I c) A d) A^2

6. If the matrix $\begin{bmatrix} 1 & 1-x \\ 2 & 3-x \end{bmatrix}$ is not invertible, then x is
 a) -1 b) 1 c) -2 d) 3
7. The value of $\sin(\sin^{-1}x + \cos^{-1}x)$ is
 a) -1 b) 0 c) 1 d) $\pi/2$
8. Two straight lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are coincident (overlapping) if
 a) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ b) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
 c) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ d) $\frac{a_1}{a_2} \times \frac{b_1}{b_2} = -1$.
9. The points which lie either side of a line $x + y = 5$ are
 a) $(0, 5)$ & $(5, 0)$ b) $(6, 1)$ & $(1, 6)$
 c) $(6, 1)$ & $(2, 2)$ d) $(2, 2)$ & $(1, 1)$
10. If $f(x) = \begin{cases} 2x + 3 & \text{for } x < 2 \\ -x - 5 & \text{for } x \geq 2 \end{cases}$, then $\lim_{x \rightarrow 2} f(x)$ is
 a) -7 b) 7 c) 1 d) not exist
11. The derivative of $f(x) = \cos x$ at $x = \frac{\pi}{3}$ is
 a) $\frac{\sqrt{3}}{2}$ b) $-\frac{\sqrt{3}}{2}$ c) $\frac{1}{2}$ d) $-\frac{1}{2}$

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) Define tautology. Prove that $p \vee (\sim p)$ is a tautology. [2]
 b) If A, B and C are the subsets of universal set U , prove that [3]
 $A - (B \cup C) = (A - B) \cap (A - C)$.

13. a) A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 3x + 1, x \in \mathbb{R}$.
Show that f is bijective. Also, obtain the formula for f^{-1} . [3]
b) Find domain and range of a function $f(x) = -x^2 + 6x - 1$. [2]
14. a) Show that the matrices $A = \begin{bmatrix} -3 & -2 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix}$ are inverse of each other. [2]
b) Prove that : $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+x \end{vmatrix} = x^2(x+3)$ [3]
15. a) Solve for the general solution of $x: \cos x + \sin x = \sqrt{2}$ [2]
b) Prove that: $2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ [3]
16. a) Find the distance between parallel lines $3x + 2y - 7 = 0$ and $6x + 4y + 9 = 0$. [2]
b) If p is the length of perpendicular dropped from the origin on the line whose x and y intercepts are a and b respectively, then prove that $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$. [3]
17. If P_1 and P_2 be the length of the perpendiculars drawn from two points $(\cos \theta, \sin \theta)$ and $(-\sec \theta, \operatorname{cosec} \theta)$ on a line $x \sec \theta + y \operatorname{cosec} \theta = 0$ respectively, prove that: $\frac{4}{P_1^2} - P_2^2 = 4$. [5]
18. a) Evaluate: $\lim_{x \rightarrow \infty} (\sqrt{x} - \sqrt{x-3})$ [2]
b) Evaluate: $\lim_{x \rightarrow y} \frac{\tan x - \tan y}{x-y}$ [3]

19. a) Find from first principles, the derivative of x^2 . [2]
 b) Find $\frac{dy}{dx}$ when $y = \frac{1}{\sqrt{x+a} + \sqrt{x-a}}$ [3]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) Construct the truth table for the compound statement $p \Rightarrow (\sim q)$. [2]
 b) If $A \cap B = \phi$, prove that $B \cup \bar{A} = \bar{A}$. [2]
 c) For any real number 'x' and any positive real number 'a' prove that
 $-a \leq x \leq a \Rightarrow |x| \leq a$. Using the implication, rewrite the inequality
 $-4 \leq x \leq -1$ using absolute value sign. [2+2]
21. a) Let $A = \{-1, 0, 1, 2, 3\}$, $B = \left\{-1, 0, 1, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}\right\}$ and function $f: A \rightarrow B$ be defined by $f(x) = \frac{x}{2+x}$. Find the range of f . Is the function f is one to one and onto both? If not, how can you make it one to one and onto both? [2+1+1]
 b) Let a function $f(x)$ be defined by
- $$f(x) = \begin{cases} 2 - x^2 & \text{for } x < 2 \\ 3 & \text{for } x = 2 \\ x - 4 & \text{for } x > 2 \end{cases}$$
- Verify that the limit of the function exists at $x = 2$. Is the function continuous at $x = 2$, why? State how can you make it continuous? [4]
22. a) Find from first principles, the derivative of $\sin 2x$. [4]
 b) Find $\frac{dy}{dx}$ if $xy^2 - x^2y = 4$. [4]

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